MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

SECOND SEMESTER EXAMINATION

(2024 ADMISSION ONWARDS)

MG2DSCPHY100 – Modern Physics

Duration: 1.5 hrs Maximum Marks: 50

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Type Questions

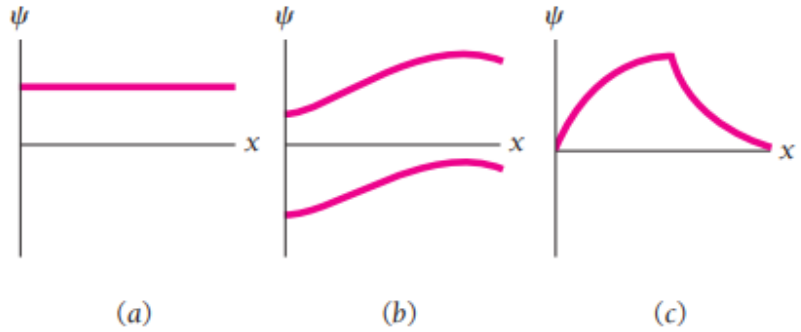
Answer any **10** questions

Each question carries **2** Mark

1	Distinguish between inertial and non-inertial frames of reference	[U][1]
2	State the two postulates of special relativity.	[R][1]
3	What will be the shape of a spherical ball moving with a relativistic velocity in the x direction as observed by an observer at rest? Explain.	[A][1]
4	Calculate the energy equivalent of 1 kg of coal.	[R][1]
5	What is meant by the work function of a metal?	[U][2]
6	In the photoelectric effect, what happens if the frequency of incident light is increased while keeping intensity constant.	[U][2]
7	What is meant by ultraviolet catastrophe?	[U][2]
8	Define matter waves. Also give the equation for wavelength of a matter wave.	[R][2]
9	What is called a wave function? What is its significance?	[R][2]
10	Distinguish between continuous spectra and line spectra.	[U][3]
11	Which series of Hydrogen spectrum lies in the visible region? write the transitions.	[U][3]
12	What is called pumping in a laser system?	[R][4]

13 Write Steady state form of Schrodinger wave equation. [R][5]

14 Given the following wavefunctions, which are not admissible wave functions? Explain. [U][5]



(2x10=20)

Part B

Short Essay Type Questions

Answer any **6** questions

Each question carries **5** marks

15 A particle moving at $0.95c$ has a proper lifetime of 2 μs. What is its observed lifetime in the laboratory frame? [U][1]

16 Calculate the length of a meter scale moving in x-direction with a velocity $0.8c$, where c is the velocity of light in vacuum. [U][1]

17 The minimum frequency for photoelectric emission in copper is 1.1×10^{15} Hz. Find the maximum energy of the photoelectrons when light of frequency 1.5×10^{15} Hz is directed on a copper surface. [U][2]

18 Derive the expression for energy density of blackbody radiation using Planck's law. [R][2]

19 Show how Heisenberg's uncertainty principle prevents an electron from existing inside a nucleus by estimating the minimum uncertainty in energy for an electron confined within a nuclear radius ($\sim 10^{-15} m$). [A][2]

20 Find the shortest and longest wavelengths in the Balmer series spectrum of Hydrogen. [U][3]

21 Expand the acronym LASER. Explain four properties of LASER light [U][4]

22 Find the probability that a particle trapped in a box L wide can be found between 0.45L and 0.55L for the ground and first excited states. [U][5]

(5x6=30)